

YEAGER, WV, USA

WMO: 724140

Lat: 38.379N

Lon: 81.590W

Elev: 910

StdP: 14.22

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13866

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.6	14.8	-1.0	5.4	14.7	3.9	6.9	18.1	20.7	41.3	18.6	42.6	3.7	210	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	91.1	72.9	88.8	72.4	86.7	71.7	76.6	85.7	75.3	83.9	74.2	82.3	6.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.0	131.1	80.3	72.7	125.4	78.8	71.6	120.8	77.7	40.5	86.2	39.2	84.2	38.1	82.2	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.9	15.3	12.5	2.1	94.5	6.6	3.5	-2.7	97.0	-6.6	99.0	-10.3	100.9	-15.1	103.4		
			1.4	79.0	6.5	1.7	-3.3	80.2	-7.1	81.3	-10.7	82.2	-15.5	83.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.0	34.5	38.0	45.8	57.0	64.5	72.1	75.1	74.3	68.1	56.9	46.2	38.7
	DBStd	16.74	12.17	11.09	10.87	9.36	7.54	5.24	4.38	4.38	6.48	8.49	9.23	10.15
	HDD50	1699	502	357	213	38	3	0	0	0	0	31	180	375
	HDD65	4385	946	756	601	269	106	9	0	1	42	275	564	816
	CDD50	3897	22	22	82	248	453	663	777	753	543	244	66	24
	CDD65	1108	0	0	5	29	91	223	312	289	135	23	1	0
	CDH74	9554	1	4	61	392	877	1888	2603	2308	1148	255	17	0
	CDH80	3213	0	0	8	86	244	634	965	828	395	52	1	0
	WSAvg	4.1	5.4	5.4	5.4	5.5	4.0	3.6	3.1	2.6	2.7	3.3	4.1	4.7
	PrecAvg	43.9	3.1	3.1	3.9	3.4	4.2	4.1	5.4	3.9	3.6	2.8	3.2	3.3
Precipitation	PrecMax	67.1	6.5	7.8	8.3	6.3	8.7	10.7	14.2	10.3	11.6	6.1	9.1	8.0
	PrecMin	29.2	1.0	0.5	1.3	0.5	0.8	0.6	2.0	0.3	1.1	0.1	0.6	0.4
	PrecStd	7.6	1.4	1.5	1.6	1.2	2.1	2.0	2.4	1.9	2.2	1.5	1.5	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.3	72.6	80.5	86.6	88.9	92.1	94.1	93.4	92.3	85.5	76.5	69.8
		MCWB	58.4	58.9	61.3	63.2	69.3	72.5	75.2	73.8	69.9	68.0	61.2	58.8
	2%	DB	63.8	65.9	74.8	82.6	85.5	88.9	90.8	90.2	88.5	80.4	71.4	64.3
		MCWB	55.7	54.6	58.4	62.0	68.5	72.4	74.6	73.2	70.4	66.1	58.3	55.5
	5%	DB	59.0	61.0	69.4	78.9	82.9	86.5	88.2	87.6	84.7	76.4	67.1	60.0
		MCWB	53.0	51.6	54.9	60.3	67.7	71.6	73.7	72.7	69.4	62.9	55.7	53.1
	10%	DB	53.6	55.8	63.9	74.5	79.4	83.9	85.6	84.9	80.9	71.9	62.6	55.5
		MCWB	47.9	48.8	52.8	58.5	66.5	70.6	72.8	72.1	67.9	61.7	53.7	49.6
	0.4%	WB	60.5	61.2	63.1	67.7	73.3	76.9	78.8	78.1	75.4	71.9	64.0	60.5
		MCDB	66.7	69.7	77.3	80.5	83.3	86.4	89.3	86.6	84.5	79.0	72.9	67.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	57.1	57.1	60.2	64.7	71.3	75.1	76.9	76.3	73.2	68.6	60.7	57.6
		MCDB	62.2	63.0	70.3	76.2	81.2	84.2	86.5	84.8	81.0	76.4	68.0	62.8
	5%	WB	53.7	53.2	57.9	62.7	69.6	73.5	75.4	75.0	71.5	65.8	57.8	54.3
		MCDB	59.0	59.2	66.6	73.5	78.7	82.7	84.1	83.2	79.1	72.5	64.4	58.7
	10%	WB	48.3	49.1	54.8	60.7	68.0	72.3	74.3	73.6	70.1	63.3	54.9	50.3
		MCDB	53.2	55.1	62.4	70.7	76.4	80.7	82.2	81.2	77.5	69.8	61.5	54.9
Mean Daily Temperature Range	5% DB	MDBR	16.4	17.7	20.3	23.4	21.4	19.9	18.9	19.5	21.0	21.7	20.3	16.5
		MCDBR	21.4	26.6	29.1	29.6	26.3	23.8	22.3	23.2	26.3	27.4	27.6	23.1
		MCWBR	14.8	18.0	15.8	13.2	11.3	9.5	8.4	8.5	10.0	13.2	16.0	15.8
	5% WB	MCDBR	19.2	23.4	23.4	22.9	21.2	19.7	18.7	19.0	20.8	21.6	21.8	19.7
		MCWBR	14.3	18.1	14.4	11.8	10.1	8.8	8.2	8.0	9.0	11.7	14.6	15.1
Clear-Sky Solar Irradiance	taub		0.307	0.322	0.347	0.389	0.436	0.460	0.488	0.491	0.401	0.360	0.325	0.300
	taud		2.495	2.442	2.429	2.323	2.194	2.172	2.114	2.079	2.324	2.439	2.513	2.540
	Ebn at Noon		278	287	290	282	269	262	253	249	268	269	269	273
	Edh at Noon		25	30	34	40	46	47	49	50	37	30	24	23
All-Sky Solar Radiation	RadAvg		569	787	1146	1526	1731	1913	1797	1692	1414	1017	723	502
	RadStd		62	113	88	122	144	111	100	89	139	127	90	61

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+270	+166
(47)	Regional (26 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+238	+119

Nomenclature: See separate page